

	reduces the force acting on the driver.
3(a)	As carriage A comes down, it loses gravitational potential energy. As carriage B goes up, it gains gravitational potential energy.
3(b)(i)	energy gained = gravitational PE gained $= 1400(9.81)(30 \sin 65^\circ)$ $= 3.7342 \times 10^5 \text{ J}$ $\approx 3.73 \times 10^5 \text{ J}$
3(b)(ii)	Loss in energy of carriage A = $800(9.81)(30 \sin 65^\circ)$ $= 2.1338 \times 10^5 \text{ J}$ $\frac{\text{loss in energy of carriage A}}{\text{gain in energy of carriage B}} = \frac{2.1338 \times 10^5}{3.7342 \times 10^5}$ $= 0.57 \text{ (2 s.f.)}$
3(b)(iii)	work done by motor = $3.73 \times 10^5 - 2.13 \times 10^5$ $= 1.60 \times 10^5 \text{ J}$ power input of motor = $\frac{1.60 \times 10^5}{60(0.6)}$ $= 4440 \text{ W}$
4(a)	Nuclear fusion is the combination of two smaller nuclei to form a larger nucleus.